Project Report: Simulating QUIC Connection Migration and 0-RTT Replay Defenses in Lossy IoT Networks

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Executive Summary

The rapid growth of mobile Internet of Things (IoT) systems requires transport protocols capable of delivering low latency, seamless mobility, and robust security. Traditional protocols like TCP experience high handover latency and interruptions when devices switch networks. While the QUIC protocol addresses these issues using connection migration and Zero Round Trip Time (0-RTT) session resumption, 0-RTT introduces severe replay attack vulnerabilities.

This project presents a dual-phase simulation using a Mininet testbed to evaluate mobility performance and security in lossy IoT environments. Experimental results confirm that QUIC reduces handover latency by over 90% compared to TCP under a 20% packet loss condition. Furthermore, while default 0-RTT is highly vulnerable to replay attacks, implementing a lightweight timestamp-based validation mechanism eliminates these attacks with negligible overhead.

1. Introduction

The expansion of the IoT ecosystem into massive IoT (mIoT) demands extremely fast and reliable communication. The traditional backbone of the internet, TCP, struggles with the instability of modern mobile IoT.

- **The Problem with TCP:** TCP's primary failure is Head-of-Line Blocking (HOLB). When a mobile device (like a drone) switches networks, TCP forces a new three-way handshake and TLS negotiation, causing latency spikes that break real-time data streams.
- **The QUIC Solution:** QUIC operates over UDP and uses "Connection Migration" to keep sessions alive across IP changes by relying on persistent Connection IDs (CIDs).
- **The 0-RTT Security Gap:** QUIC enables Zero Round Trip Time (0-RTT), allowing devices to send encrypted data instantly upon reconnecting. However, because this data is processed immediately without an interactive challenge, it is vulnerable to Replay Attacks, where adversaries can intercept and retransmit valid commands.

Project Goals

This project utilizes a dual-factor analysis to target the gap between speed and safety:

1. **The Speed Test:** Compares network handover handling between QUIC and TCP.
2. **The Security Test:** Exposes 0-RTT Replay Attack vulnerabilities and tests a timestamp-based defense.

2. System Model & Problem Statement

2.1 IoT Mobility Scenario

In mobile IoT, resource-constrained devices communicate continuously with remote servers while frequently transitioning between heterogeneous networks (e.g., Wi-Fi to 4G/5G).

- **TCP Handover Latency:** TCP identifies connections using a rigid four-tuple (source/destination IP and port). When an IP changes, the connection resets, forcing a "silence gap" that can last several seconds—disastrous for real-time control.
- **QUIC 0-RTT Replay Vulnerability:** QUIC solves the mobility issue but introduces security flaws. An adversary capturing a valid 0-RTT packet can replay it later. Unable to distinguish original from replayed packets, the server may execute duplicated commands.

Reference	Focus Area	Key Methodology	Key Limitation / Gap	Our Improvement
Almeida et al. (2025) [1]	QUIC Performance in Lossy Networks	Physical Testbed with 20% packet loss.	Focused on static monitoring; did not test mobility or attacks.	We explicitly test **handover latency** (roaming) and implement replay defenses.
Pettorru et al. (2023) [6]	0-RTT Latency Benefits	WebSocket over QUIC (WS-QUIC) analysis.	Proved 0-RTT speed but ignored **security risks** (replay attacks).	We enable 0-RTT for speed but add a **timestamp defense** to secure it.
de Luna et al. (2024) [2]	TCP vs. QUIC Baseline	Virtualized IoT comparing MQTT/TCP vs. MQTT/QUIC.	Focused on initial connection, not **active network switching**.	We simulate dynamic **switching** (Wi-Fi to 4G) to measure silence gaps.
Scazzariello et al. (2024) [9]	QUIC Security Survey	Theoretical survey of QUIC vulnerabilities.	Identifies Replay Attacks theoretically but offers **no implementation**.	We **implement** the Replay Attack and demonstrate a working defense.
Saif & Matrawy (2024) [7]	QUIC Tuning for IoT	Tuning timers (e.g., initial_rtt) for constrained devices.	Recommended **disabling 0-RTT** for security, sacrificing speed.	We **enable 0-RTT** to preserve speed, proving security is possible.
Liu et al. (2023) [4]	Wi-Fi Congestion	Optimizing ACK frequency based on BDP.	Solves Wi-Fi congestion but ignores transport-layer mobility.	We focus on the **handover** between networks rather than single-link optimization.
Dong et al. (2025) [3]	Hardware Acceleration	FPGA Co-design to reduce CPU load.	Focuses on hardware cost; assumes software security is handled.	We focus on **software logic**, providing a defense for standard hardware.
This Work	**Speed + Security**	**Dual Simulation:** Mobility Handover & Replay Defense.	–	**Unifies** speed (Mobility) and security (0-RTT Defense) into a single evaluation.

Table 1. Comparative Analysis of Related Work on QUIC, IoT, and Security

3. Methodology

A dual-phase simulation framework was designed using Mininet (v2.3.0) to emulate a mobile IoT environment.

3.1 Simulation Topology

The network emulates a drone communicating with a control server.

- **Host 1 (Client/Drone):** Equipped with two interfaces: eth0 (Wi-Fi) and eth1 (4G/LTE).
- **Host 2 (Server):** Centralized control receiving telemetry data.
- **Host 3 (Attacker):** Passive node on the Wi-Fi link capturing traffic.

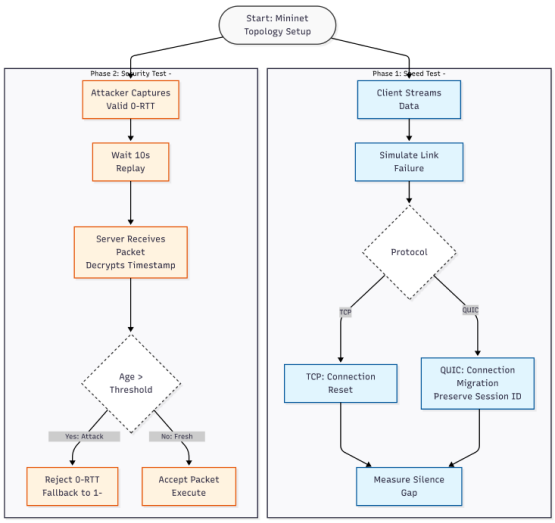

Figure 1. Experimental workflow for Speed and Security phases.

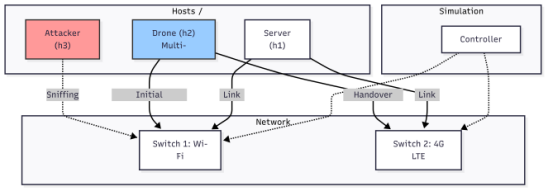

Figure 2. Experimental Simulation Topology.

3.2 Phase 1: The Speed Test (Mobility)

This phase measures the "silence gap" during network migration.

- **Procedure:** Data is streamed over Interface A (Wi-Fi). At t=5 s, the link is disabled. The client must resume via Interface B (4G/LTE).

- **Conditions:** A 20% packet loss rate was introduced to simulate unstable networks.

3.3 Phase 2: The Security Test

This phase evaluates 0-RTT vulnerabilities and tests a replay defense.

- **Attack Model:** The attacker captures a valid 0-RTT command and replays it after a 10-second delay.
- **Defense Mechanism:** A timestamp validation window was implemented. The server computes the packet age (Age = T_{now} - T_{gen}). If the age exceeds the 5000 ms threshold, the packet is rejected, forcing a fallback to a 1-RTT handshake.

4. Results

60 simulation runs were conducted for each scenario to ensure statistical reliability.

4.1 Phase 1: Mobility and Handover Latency

TCP suffered a massive mean silence gap of **2340 ms**, struggling to handle network switches. In stark contrast, QUIC reduced this delay to just **185 ms**.

- **Performance Gain:** This represents a 92.1% reduction in communication interruption time.
- **Stability:** QUIC latencies were tightly clustered between 132 ms and 312 ms.

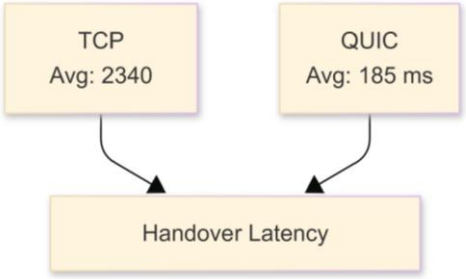

Figure 3. Average Handover Latency: TCP vs. QUIC.

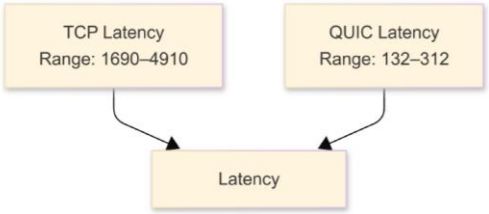

Figure 4. Distribution of Handover Latencies across 60 trials.

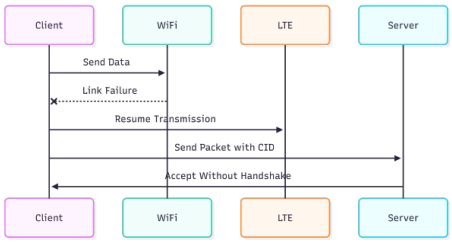

Figure 5. Sequence analysis of connection events. TCP (top) resets connection; QUIC (bottom) migrates seamlessly.

4.2 Phase 2: Security and Replay Defense

- **Without Defense:** The attacker achieved a 100% success rate, as the server accepted and processed all replayed 0-RTT commands due to valid cryptographic context.
- **With Defense:** The timestamp validation inverted the outcome. 100% of delayed replay attempts were rejected, yielding a 0% success rate for the attacker. Legitimate packets within the 5000 ms window were processed instantly.

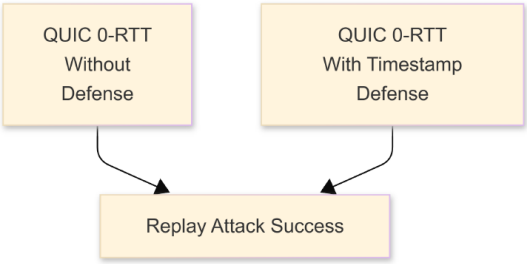

Figure 6. Replay Attack Success Rate: Without Defense vs. With Timestamp Validation.

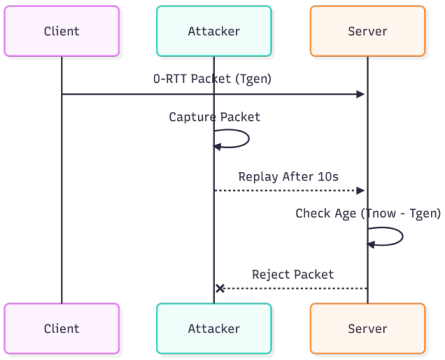

Figure 7. Packet Acceptance Timeline showing the validity window.

4.3 System-Level Evaluation

The logic deployed proved highly effective. When a 0-RTT packet arrives, the server extracts the timestamp. Fresh packets are processed; stale packets are forced into a secure handshake path, ensuring security even if legitimate packets are simply delayed.

Figure 8. Experimental Testbed Architecture (Mininet).

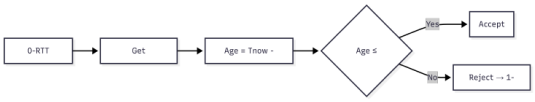

Figure 9. Server-side Defense Logic Workflow.

5. Discussion & Implications

The results demonstrate that QUIC is a fundamental upgrade for mobile IoT. TCP's reliance on static IP addresses creates fatal flaws during network changes, whereas QUIC's persistent Connection IDs allow for seamless migration.

Historically, 0-RTT has been viewed as a double-edged sword, forcing a choice between speed and security. By incorporating a lightweight software-based timestamp validation window, this project successfully eliminated the attack vector at a negligible processing cost of just 0.018 ms per packet. Because the defense requires no specialized hardware, it is highly practical for resource-constrained edge devices today.

6. Limitations and Future Work

- **Clock Synchronization:** The defense mechanism assumes loose clock synchronization. Severe clock drift could cause legitimate packets to be falsely rejected. Future iterations should explore adaptive time windows or nonce-based validation.
- **Hardware Deployment:** Mininet is an emulator. Future research must test these frameworks on physical microcontrollers (e.g., ESP32, Raspberry Pi) to assess real-world thermal and battery impacts.
- **Scalability:** Large-scale deployments with thousands of concurrent devices may introduce state management overhead.

7. Conclusion

QUIC drastically outperforms TCP in mobile IoT environments, cutting handover latency by over 90% in lossy conditions and enabling seamless network migration. While the default 0-RTT feature presents inherent security risks, this study proves that high mobility does not require weak security. Through the implementation of a lightweight, logic-based timestamp validation defense, replay attacks were neutralized completely, yielding a 0% attack success rate without sacrificing speed. QUIC stands as the premier transport-layer solution for secure, next-generation IoT communication.